# 建構感知親生命性設計於環境體驗之概念架構

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**【摘要】**

親生命假說(Biophilia Hypothesis)是指人類具有親近自然及生命體的本能，而親生命性設計強調將自然融入建成環境，進而促進生心理健康之效益。上述的論述呼應環境心理學及東方傳統環境氣等探討環境屬性、特徵與人互動關係之概念。目前親生命性設計僅著重於概念上，尚未發展評估項目，故本研究從親生命性設計理論，試圖精準地理解概念與含義後，採用專家小組、景觀專業者及非專業者語意及用字校正，將Kellert、Heerwagen 與 Mador (2008)提出72項環境特徵與元素萃取57項，建構59題評估建成環境中感知親生命性設計的項目。這項研究的貢獻為提出感知親生命性設計項目，測量人感受到之環境屬性、特徵與元素及其體驗，藉由感知體驗對環境評估的方式，能應用在實質景觀與環境規劃及設計方面。

**關鍵詞：**感知親生命性設計項目、環境屬性、環境特徵、體驗與效益、 建成環境

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# Building Conceptual Framework of Perceived Biophilic Design on Environmental Experiences

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## Abstract

The Biophilia Hypothesis proposes that people have the instinct to be with nature and living bodies. Using the biophilic design can integrate nature into the built environment, thereby promoting health benefits such as physical and psychological health benefits, which is in line with related environmental psychology theories and Traditional Environmental Qi. However, there have not yet been developed evaluation items that can measure perceived biophilic design. The research builds on biophilic design theory to accurately understanding the concept and meanings and cooperates with expert group discussions, landscape professionals, and non-professionals to adjust semantics and the use of words in questionnaires. The research extracts 57 environmental features and elements from 72 items, which produced by Kellert, Heerwagen, & Mador (2008) and develops 59 questions to evaluate the items of perceived biophilic design in the built environment. This research contributes to providing a Perceived Biophilic Design Items to measure a sense of experience in nature. Through the overall perception evaluation method, it can be applied to the planning and design in the landscape.

**Keywords** ： Perceived Biophilic Design Items, Environmental Attributes, Environmental Features, Experiences and Benefits, Built Environment

# 壹、前言

人與自然互動經驗(people-nature experiences)最早由美國社會心理學家Eric Fromm (1964)在《人心及其善惡本性(The Heart of Man: Its Genius For Good and Evil)》的書中提到親生命(biophilia)與親生命性(biophilic)，他認為「親生命」指熱愛生活(the love of life)：非由單一特徵而是整體性的特徵，呈現個體的情緒、思想並且能被生活與成長的過程吸引；「親生命性」有其善惡倫理原則：服務生命視為善；服務死亡視為惡。在1984年生物學家Wilson依照Fromm的概念，提出親生命假說(Biophilia Hypothesis)，認為人類與自然界有著深厚的淵源並植根於基因中，並用其解釋人與環境之間的連結感。Kellert、Heerwagen和Mador (2008)《親生命性設計：理論、科學與建築物栩栩如生的實踐(Biophilic Design: The Theory, Science and Practice of Bringing Buildings to Life)》的書中提到親生命性設計(Biophilic design)，試圖以環境屬性與其包含之環境特徵與元素設計於建成環境，促進人與建成環境產生良好互動模式，進而獲得生心理健康的效益。

親生命性設計的概念是將具有潛力的自然特徵或元素整合到建成環境中，提供人們接觸到自然、獲得生心理健康效益；此概念與環境心理學及景觀相關理論相互呼應，例如：Appleton(1975)提出有利於生存的環境條件，是人們能觀察到前景(Prospect)而不被看見(Refuge)，與景觀組成、形狀、顏色或配置有關；稀樹草原假說(Savanna Hypothesis)提到人們喜歡非洲大草原的地形，一望無際、具開闊性與配置一株樹木或些許灌木組成的景觀(Balling & Falk, 1982)。上述的論述與棲地理論(Habitat theory)及景觀偏好矩陣 (Preference Matrix)有異曲同工之妙，皆認為選擇棲息地的原則為能夠探索該地的資源、避免資源匱乏外(Orians & Heerwagen, 1992)，必須能理解環境所提供的訊息、探索與推論，進而預測景觀偏好(Kaplan & Kaplan, 1989)。

近年來，國際期刊與相關研討會也逐漸重視親生命(biophilia)與親生命性設計(biophilic design)的議題(Browning, Ryan, & Clancy, 2014; Clancy & Ryan, 2015; Downton, Jones, Zeunert, & Roos, 2017)，探討如何應用親生命性設計手法於日常生活環境中，例如：在市區道路景觀增添行道樹、都市公園、綠建築物環境、或在辦公室、學校、醫院等環境內擺設植栽或照片 。另從Kellert等人(2008)和Kellert(2018)書中的案例，可看出親生命性設計的原則、環境屬性及特徵，皆會影響生心理健康的概念。

從親生命性設計的概念與生心理效益的文獻回顧中發現，採用親生命性設計的屬性（視覺與非視覺接觸自然、自然照片、自然材質、植栽等）能增進生心理效益（減緩壓力、注意力恢復、增進正向情緒、偏好等），並從文獻回顧也指出此概念與恢復性環境、景觀偏好、壓力減輕理論等具相關性(Hung & Chang, 2021)。然而，親生命性設計尚停留在概念層次或僅採用實驗設計手法探討，目前尚無量表具體測量人感知到的環境，究竟涵蓋哪些親生命性設計的特徵與元素是會影響生心理體驗。另一方面，相較於自然連結量表、環境態度量表、知覺恢復性量表等，本研究提出「感知親生命性設計項目」著重從個人感官知覺與經驗，體驗環境的景觀元素（水、植栽、光影、有機物的生長與變化等）、組成與佈局（自然形狀與樣式之配置、塑造空間大小）等之設計手法，進而評價整體環境與人互動過程中所產生之心理感受。因此，本研究試著探索都市綠地中，具親生命性設計的景觀建成環境，例如環境特徵、自然形狀與樣式、自然形式與過程而非涵蓋人造設施物等之景觀設計手法，其環境是否能自然而然吸引人們的注意、引發探索與好奇心。故本研究目的為重新檢視Kellert等人(2008)提出親生命設計元素與特徵的概念意義，界定研究範疇為建成環境下之戶外都市綠地，將環境中各項特徵與元素具體化，提出未來可實際量測其概念意義的項目。研究貢獻期許未來該親生命性設計之概念架構，能作為實務規劃與設計上衡量人感知親生命性設計特徵與元素與自然體驗之參考依據。

# 貳、文獻回顧

## 一、人與環境互動模式及其心理反應

人與自然互動經驗是探討個體透過感官知覺、體驗自然與環境後並從中獲得生心理健康效益的互動模式，這樣的模式與親生命假說的概念雷同。親生命假說認為人對自然有種與生俱來的親密感(“humans have a deep affiliation to nature that is rooted in our own biology”)進而產生關注生命或像生命的過程(“the innate tendency to focus on life and life-like processes”)，這種親密感會呈現在與自然相關的活動或環境中(Wilson, 1984)。Keniger、Gaston、Irvine與Fuller (2013)的研究以「自然(nature)」、「互動(interaction)」、「效益(benefits)」、「健康(health)」、「生態多樣性(biodiversity)」等的關鍵字，探討人與自然互動(interactions between people and nature)的模式，歸納出人與自然互動的場域有室內建築環境、都市環境、近郊（或邊緣景觀）、生產景觀、荒野以及其他等六種環境；互動類型分為三種：第一種是無直接接觸，意旨體驗自然時個體並沒有直接「在」自然環境中，例如：透過窗景觀賞景觀；第二種為間接接觸，指將自然視為活動的附屬品，例如：在上班、上學途中無意間看到自然景色；第三種為直接地接觸自然並且在環境中從事活動，如野餐、露營等；互動所帶來之效益可分為生心理及認知，例如：增加自我認同/幸福感、提升正向情緒、降低挫敗感/焦慮感/壓力/血壓、提升注意力恢復/工作表現/生產率等。這三種與自然互動的模式經常出現在每日生活的環境中，能無形中影響人們的生心理反應。另一方面，過去大量的實證研究皆不斷證實接觸自然、綠環境或元素比起高樓林立的都市與灰色基礎設施所組成的環境，更具有釋放壓力、恢復注意力、提升正向情緒等生心理健康效益(Bratman, Daily, Levy, & Gross, 2015; Hartig, Evans, Jamner, Davis, & Gärling, 2003; Kaplan & Kaplan, 1989; Meredith et al., 2020; Tsunetsugu et al., 2013; Ulrich et al., 1991)。

親生命假說有助於解釋人們尋求與自然接觸的益處。Kellert (1996)依據Wilison (1984)提出親生命傾向之九種價值指標，分別為: 「功利主義(utilitarian)」、「自然主義 (naturalistic)」、「生態主義科學(ecologistic scientific)」、「美學(aesthetic)」、「象徵(symbolic)」、「人文(humanistic)」、「道德(moralistic)」、「支配(dominionistic)」及「消極感(negativistic)」等作為與自然連結的概念；Delavari-Edalat與Abdi (2010)運用上述概念擬定量表並結合質性訪談，探討人們對都市公園提供人與環境互動的親生命體驗價值。Mayer與Frantz (2004)提出個人對自然連結的特徵量表(Trait of Connectedness to nature scale, CNS)，主要量測與自然界的一體感(sense of oneness with the natural world)、與動植物的親緣關係(sense of kinship with animals and plants)及自我與自然的平等感(sense of equality between self and nature)等人與自然情感上的連結；進一步於2009年將特徵量表轉換成詢問當下人與自然互動關係的「狀態量表」，測量受測者當下行走於自然環境中，與自然環境互動後所產生注意力恢復、正向情緒、反思自我生活等之心理效益(Mayer, Frantz, Bruehlman-Senecal, & Dolliver, 2009)。然而，上述量表僅探討人與自然互動之情感連結，尚未以景觀與環境設計的觀點切入並了解人感知環境中的景觀特徵、元素與其相關環境設計手法及其引發之心理感受。另一方面，有研究探討人對生態行為(ecological behavior)、環境態度(environmental attitude) (Collado, Corraliza, Staats, & Ruiz, 2015; Sandifer, Sutton-Grier, & Ward, 2015)之議題，如採用個人對生態看法(Ecology Scale) (Maloney & Ward, 1973)、環境關注(Environmental Concern Scale) (Weigel & Weigel, 1978)或新環境典範(The new environmental paradigm, NEP) (Dunlap & Van Liere, 1978)等之環境態度與自然開發或保護相關量表。上述眾多的量表著重探討對環境的態度、行為對環境情感上連結的心理狀態，尚無法涵蓋Kellert等人所提出之「親生命性設計」概念，並且有別於本研究重視人與環境互動下，人是否能感知環境中的景觀屬性、特徵、元素與其心理體驗。因此，本研究有必要重新建構「感知親生命性設計」應用於建成環境之架構。

從西方環境心理學的角度探討人與環境互動的研究，皆已不斷證實相對於都市環境，自然環境能引發出正向情緒與生心理健康效益，其相關的理論有注意力恢復理論、壓力削減理論、偏好矩陣等。注意力恢復理論視環境中具有恢復持續使用直接注意力而產生疲勞的特徵－遠離性(Being away)、魅力性(Fascination)、延展性(Extent)與相容性(Compatibility)等(Kaplan & Kaplan, 1989)；壓力削減理論則是探討當個體處於具有壓力情境中，藉由接觸或觀看自然景觀/元素來調節情緒及恢復生理上處於緊張或高壓的狀態(Ulrich, 1984)；又個體透過視覺接受到環境刺激後，所產生之心理評價稱之為景觀偏好，其影響偏好之維度又可分為「理解與探索」與「推斷與立即」分別由四項影響因子組成－一致性(Coherence)、複雜性(Complexity)、易讀性(Legibility)與神秘性(Mystery) (Kaplan, 1987)。

從眾多的社會科學研究，不論是直接接觸或間接接觸自然景觀、都市自然、都市景觀等，實驗結果不斷驗證自然環境或者有自然元素的環境，能恢復直接注意力、釋放壓力與提升正向情緒(Hartig, Mang, & Evans, 1991; Ulrich et al., 1991) 並且相較於都市街區，人們對都市自然的偏好高且其一致性、複雜性與神秘性等因子，能預測景觀偏好(Abkar, Kamal, Maulan, & Davoodi, 2011)。然而，Velarde、Fry和Tveit (2007)的回顧型研究提到透過視覺刺激，廣泛性地分類自然與都市環境所產生之效益，僅能粗略討論相較於都市環境，自然環境確實能提供影響生心理與情緒健康效益，但無法明確指出哪些「關鍵元素」能提供較佳益處。Van den Berg、Jorgensen和Wilson (2014)的研究中也提到都市環境中的綠地、樹林與野生林地等的景觀類型對情緒和恢復體驗無顯著影響，然而綠環境在心理反應上皆對都市建築且無綠化的環境有明顯差異。綜上所論關於景觀評估為單一或部分關鍵因子或是其因子間的組合形式，尚須許多未來研究針對不同實驗設計與面項加以評估。本研究僅從親生命性設計的概念，探討環境中之景觀或特徵所帶出之整體氛圍，而非探討人造設施物或量體樣式，故以人對體驗環境的角度切入，作為後續評估人與環境互動模式及其心理效益之研究操作趨勢與方向。

當體驗自然時，人們需要身入其境才能獲得健康效益，而上述測量人與環境互動的觀點著重在互動行為下產生的心理狀態且大多數研究採用注意力恢復量表測量具有恢復性的環境特徵或廣泛性比較環境類型對心理效益的影響，鮮少從親生命假說與其設計概念切入。因此，本研究目的以環境心理學相關理論為基礎，連結Kellert等人 (2008)提出親生命性的設計屬性與元素作為理論基礎，逐步檢視概念、屬性與元素後，建構感知親生命性設計的項目(Perceived Biophilic Design Scale Items)。

## 二、西方親生命性設計與東方傳統環境氣之相關研究

當人們在戶外綠地環境中練氣功時，當下感受環境體驗中的健康能量，稱之為練氣體驗，與心流體驗(Csikszentmihalyi, 1990)在概念及統計檢定上具高度的相關性(Hung, Hwang, & Chang, 2021)。換句話說，從事氣功活動體驗到「氣感、天人合一、精氣神充足」的感覺時，與心流體驗之「忘我、時間流逝感有別於平常、樂趣」等的心理狀態的感受相似。進一步，從東方與西方環境觀探討人與環境互動下產生的環境體驗模型，發現具魅力性、相容性等的戶外景觀環境，能促進快樂狀態與體驗的產生(McDonald, Wearing, & Ponting, 2009; Williams & Harvey, 2001)，並且在戶外環境從事攀岩、爬山等的活動時，雖挑戰與環境不一定有關，然而當戶外活動克服環境挑戰與平衡自身技巧時，可能進入心流體驗(Csikszentmihalyi, 1990)。另一方面，運用親生命性設計特徵與元素的概念，在都市綠地如水池、樹林地、草地等環境從事氣功活動時，能體驗到心流體驗、練氣體驗、恢復體驗等心理狀態(Hung, Chou, & Chang, 2021)。因此，本研究將更進一步了解不同環境所組成的特徵與元素，究竟如何影響人與環境互動之心理反應。以下從西方環境觀–親生命性設計與東方環境觀–傳統環境氣分別論述。

### （一） 親生命性設計

景觀設計試圖運用元素、地形、地貌與環境將「自然」帶入人們生活空間，又「自然」具有無法言喻的吸引力，讓人們在體驗環境時能獲得生心理健康效益。常用的景觀設計元素包含各式各樣、形色質等的植栽、鋪面、水與結構物等，詮釋不同視野及空間配置對心理反應的感受，尤其是在建成環境中創造與「自然」的連結。親生命性設計(Biophilic design)的概念是運用自然界的特徵與元素，特別是提供生活在建成環境中的人們舒適、安全、健康等的心理感受(Kellert, 2018)。Kellert等人(2008)的書中提到親生命性設計包括兩個主要維度：有機或自然維度(Organic or Naturalistic dimension)，討論與環境互動的行為模式；地方或風土有關的維度(Place-based or Vernacular dimension) ，探討人們對當地環境的文化、生態與建成環境的情感。藉由上述自然與情感之兩項維度，組成六種親生命性設計屬性，分別為：建成環境中的環境特徵(Environmental features)包含水、空氣、陽光、動植物等自然元素；自然形狀與樣式(Natural shapes and forms)為使用動物圖案、貝殼和螺旋、拱頂、圓頂等模擬自然特徵或生物形態之室內與戶外建成環境之外觀設計與環境氛圍；自然形式與變化過程(Natural patterns and processes)主要含有多變化的環境資訊以刺激感官，例如：時間推移或老化作用、季節變化與生長、空間變化等；光與空間(Light and space)涵蓋許多光源樣式以模擬自然光影變化及空間設計，且該屬性多半與室內和建成環境設計有關；地方感與連結的關係(Place-based relationship)與地方連結感、地方精神有關；環境與人互動關係(Evolved human-nature relationship)與人對自然演化有關，例如:環境中產生眺曬、神秘感、感官知覺、探索、安全感等情感上與環境互動的關係。以上六種親生命性設計屬性涵蓋72項環境特徵與元素。

Browning等人(2014)提出14種親生命性設計模式能實質創造親近自然環境的設計，例如：以視覺接觸自然或採用非視覺接觸自然方式、提供溫溼度、氣流、光線充足等良好的舒適度佳的環境、或在環境中使用「水、植栽、或其他元素」可增添複雜度、藏匿、神祕性等有秩序的排列組合，提供不同空間設計手法增添視覺偏好、情感連結與刺激不同感官知覺，進而影響生心理效益(Ryan, Browning, Clancy, Andrews, & Kallianpurkar, 2014)。一篇2015年的研究，討論親生命性設計與健康及幸福感效益，結果指出直接地接觸自然，如乾淨的水元素或水景、自然光、氣候、植栽等或者透過窗景，觀看屋頂綠化、自然、生態或花園等，能獲得注意力恢復；在辦公室的室內環境觀看自然影像，也能恢復心理上的疲勞；然而較少的研究證據說明自然材質或幾何圖形能對心理上產生恢復性效益(Gillis & Gatersleben, 2015)。不過，也有研究指出景觀偏好與自然景觀照片（包含大量樹木與水體）的碎形維度呈現倒U字型關係（謝孟倫 & 林晏州，2011），故推論除了自然特徵外，元素與特徵所組成的複雜度也具其影響。

綜上相關研究，發現親生命性設計與環境心理學相關理論有關，例如：注意力恢復、壓力削減理論、視覺偏好、眺曬理論等，能輔助後續研究討論親生命性屬性與特徵之組合與配置，並能結合應用在建築、景觀、心理與健康等相關領域，增加親生命性之實質應用價值。然而，目前尚無研究運用親生命性設計概念轉換成心理問項，實質測量人是否能感受具有親生命性屬性與特徵環境，從中體驗與環境之親密感與其產生之生心理反應。

### （二）東方傳統環境氣

傳統環境氣的概念源自於東方環境觀，認為人與環境為一體，也就是所謂的萬物靜觀皆自得、天人合一、與自然達到平衡及和諧的狀態。該概念能連結到景觀生態學探討空間組成與佈局，例如：採用景觀結構指數代表整體空間之組成、佈局、元素與連結度等之指標，用以討論組成的數量、類型與範圍比例等；採用景觀元素的空間特徵，描述元素在空間中的排列位置、方向，例如：池塘間的距離、森林塊區的形狀與複雜度等的配置(Leitão, Miller, Ahern, & McGarigal, 2012)。

周宛俞、洪詩涵與張俊彥(2020)認為相較於一般人，資深心身運動者對環境體驗的感受較強

烈、敏銳且能體驗到環境氣與不同環境的特徵，並發展傳統環境氣(Traditional Environmental Qi, TEQ)量表。該量表以東方環境觀切入，提出環境特徵有景觀空間結構(landscape structure)、植栽特徵與生長狀態(vegetation characteristics)、微氣候(microclimate)、視覺品質(visual quality)、明亮(brightness)、干擾物(disturbances)及個人健康感受(healthy feelings)等，作為評估人與環境互動時產生平衡與和諧狀態，並與環境心理學相關理論（景觀效益與偏好、親生命假說、眺曬理論）相互呼應。舉例來說，人喜歡處在植栽配置乾淨、整齊且光線充足的環境內，感受到景觀與元素的協調性、空間尺度適當性及多樣性的自然元素等，並能體驗舒適感、安全感與放鬆。該份量表源自於探討從事心身運動與環境互動的關係，但也已使用在評估一般人對環境的體驗，其研究發現心身運動者與一般人對環境中的傳統環境氣體驗、環境恢復性體驗與偏好的感受程度不同，顯示心身運動者對環境整體上組成與配置的需求標準高於一般人（周宛俞，2016）。進一步，將傳統環境氣的概念運用在景觀生態的角度，將都市綠地環境數化，發現一般人偏好大面積的草地與整齊的樹林，然而心身運動者則喜歡在具有多樣化、不規則分布的植栽材料、大面積且連續性的水與平整的人工鋪面的環境(Chou, Lee, & Chang, 2016)。

探討人與環境互動過程中環境特徵如何影響生心理反應，能從東方練氣者的角度與西方親生命性設計切入。東方傳統環境氣量表的建構是從練氣者評估現地環境，找出影響練氣體驗的環境特徵；西方親生命性設計提出運用環境特徵與元素在建成環境中增添「自然」，進而增進親近自然的機會、提升生心理健康效益。兩者在概念上皆重視景觀特徵與元素之組成佈局對環境的體驗，並能從相關研究與環境心理學之理論詮釋。然而，Kellert等人(2008)提出72親生命性環境特徵與元素，尚需進一步釐清與建構，方能作為評估一般大眾對不同建成環境類型之親生命性設計感受之問項。另進一步建構其量表後，能比較東方與西方所重視之環境特徵或元素對環境體驗及其產生之生心理健康與效益。

## 參、研究方法

### 一、親生命性設計問項之操作流程

在2008年的書中提到六項親生命性設計屬性與元素分別為：環境特徵、自然形狀與樣式、自然形式與變化過程、光與空間、地方感與連結的關係與環境與人互動關係等，包含72項環境特徵與元素(Kellert, et al., 2008)；2018年的書中整合六項環境屬性為三項，分別為直接體驗自然(Direct experience of nature)、間接體驗自然(Indirect experience of nature)以及空間與地方(Space and place)，包含25項環境特徵與元素(Kellert, 2018)，兩者皆可作為應用在實質建成環境之設計參考依據。本研究評估絕大部分的25項環境特徵與元素包含在72項內，並經由深入討論每項親生命性設計之屬性、特徵與元素所闡述的概念與其應用後，發現72項環境特徵與元素更能豐富地描述建成環境中的細節及自然及環境互動體驗下所產生之心理感受，故本研究以評估72項特徵與元素為主。

本研究擬定親生命性設計項目的目的是在測量戶外建成環境中，人感知到親生命性特徵與元素之程度，作為衡量該環境是否具有提供親近自然的潛力，進而影響環境體驗與生心理反應之健康效益。本研究納入72項環境特徵與元素之原則為：1. 廣泛提及建成環境或景觀、2. 與景觀與建成環境相關的概念、3. 與戶外建成環境相關元素、4. 環境與人互動產生之情感。然而尚有其特徵與元素屬描述室內、裝飾或建築師設計等，故排除標準為：1. 無提及建成環境景觀或元素、2. 過於強調建築物外觀設計（人造物、設施物、裝飾品等）、室內環境模擬自然設計等相關含意與用字（例如：內部環境、建築設計、人為操縱）。接續研究進行專家小組會議、景觀非景觀人語意及用字理解度等評估過程，

如圖1研究流程圖所示。

首先，研究者先行篩選元素並進行語意翻譯、萃取關鍵字、整合屬性、特徵或元素等內容分析（表1），並將各項元素類別名稱及其內容符合環境氛圍之「概念意義」轉換成問項（表2）。舉例來說，「顏色(color)」：依據原文描述元素之環境氛圍與概念意義為「環境色彩、自然色彩、大地色彩」，問項為「環境中有自然色彩」的感受程度；「植物圖案(Botanical motifs)」其概念意義為環境中呈現或模擬「自然樣貌」之設計，讓整體環境增添植物形狀、樣式之自然體驗，轉換為問項則為「環境有模擬植物型態的設計」的感受程度。在概念意義轉換成擬定問項的過程中，進行景觀專家小組討論，分別邀請三位受過景觀專業訓練含六年以上及英文程度中高級者參與。第一次專家小組討論會議目的是逐項檢討與調整篩選後之元素，並針對10項有疑慮的元素（例如：景觀生態、地方感與連結關係之相關特徵、掌控與控制等）與元素在原屬性間之調整作討論，舉例來說表1屬性3之元素「感官豐富(sensory variability)」偏重運用感官知覺感受環境刺激，屬環境與人互動的體驗，故調整該元素至表2新屬性5「環境與人互動關係」。第一次專家小組討論，直到每位專家與研究者對上述概念內容與元素分析達一致性，則停止討論。接著，研究者將上述問項邀請10位景觀相關人士（至少受過景觀專業訓練四年以上）進行兩次語意及用字評估，第一次語意評估其理解問項的一致性率為72.88%；經與第二次專家小組討論與校正後，第二次語意評估其理解問項的一致性率達88.14%；另進一步邀請20位非景觀相關人士（男女各半，年齡20-60歲）評估校正後之問項，發現共有35題有語意、用字不清楚之疑慮。然而，經由細部分析發現，僅有3題分別有10至13位非景觀相關人士認為「立面綠化、自然線條與仿生性設計」等的用字難以理解，建議應舉例說明其概念或設計手法；其餘大部分有疑慮之問項，分別僅有1到5位非專業者人士反應語意不清或建議舉例說明。

綜合上述景觀與非景觀人士針對語意及用字之評估，本研究再次邀請第三次專家小組討論，針對親生命性設計偏重的設計手法，再次進行語意校正及舉例說明，例如：原問項9.「這裡的環境有立面綠化」校正為「這裡的環境有立面綠化（例如：攀藤植物、植生牆）」、原問項11.「這裡的環境有樹型外觀設計」校正為「這裡的環境有模擬樹型外觀的設計（例如：樹幹）、原問項12.「這裡的環境有卵形、橢圓形或管狀設計」校正為「這裡的環境有卵形、橢圓形或管狀設計（例如：花園、噴泉）」、原問項13.「這裡的環境有自然線條」校正為「這裡的環境有自然線條（例如：蜿蜒步道）」、原問項15.「這裡的環境有仿生性設計」校正為「這裡的環境有模仿生物型態的設計」等。經由上述概念架構中問項的收斂過程，本研究提出59題戶外建成環境之感知親生命性設計問項，作為未來實證研究之評估項目。

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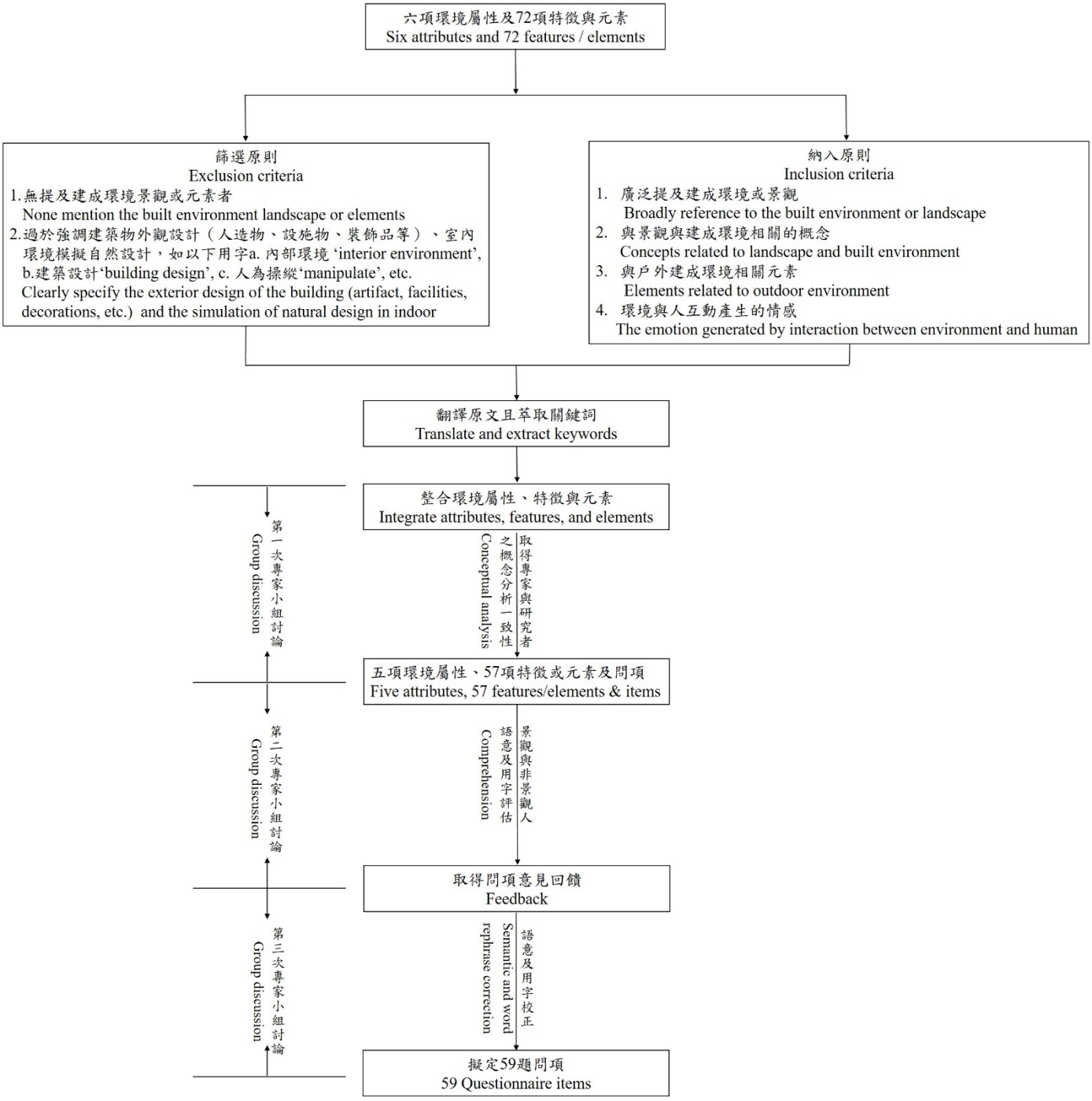


圖1　親生命性設計之環境特徵與元素篩選流程圖

## 二、整合屬性、特徵與元素並擬定問項

依據上述的環境特徵與元素篩選關鍵字、概念意義的內容分析，其評估表如表1所示。原屬性共六項，篩選後剩餘五項，而其大部分主要刪除表一屬性4「光與空間(Light and space)」與調整其剩餘特徵/元素至屬性2自然形式與過程(Natural patterns and processes)和屬性5環境與人互動關係(Evolved human-nature relationship)中。經調整後，本研究之親生命性設計涵蓋五項屬性：第一項環境特徵(Environmental features)包含顏色、水、空氣、陽光、植栽、動物、天然材質、景觀等11項特徵及元素；第二項自然形狀與樣式(Natural shapes and forms)包含使用植物形狀與樣式設計、樹形外觀設計、卵形、

自然線條、仿生性等6項特徵及元素；第三項自然形式與過程(Natural patterns and processes)包含萬物隨時間推移老化、生長、有機體與無機物間的搭配性、生物多樣性、景觀焦點、視覺延伸感、整體性、環境元素間對比性、光影對比性、重複與變化性、層次感、尺寸適當等16項特徵及元素；第四項地方感與連結的關係(Place-based relationship)包含地理、歷史、人文、生態上的連結感、地方象徵意義、無地方意義的連結感等10項特徵及元素；第五項環境與人互動關係(Evolved human-nature relationship)包含眺暨、神秘感、敬畏、吸引力、探索與發現等14項特徵及元素，其概念意義轉換成項目如表2。

表1 72項環境元素之關鍵字與內容概念分析評估表

| 屬性(A) | 元素 Elements | 節錄原文(Kellert et al., 2008) excerpt from original | 環境氛圍概念與關鍵字 concept and keywords | 評估(E) |
|---|---|---|---|---|
| 1. 環境特徵 (Environmental features) | 1. Color | Color has long been instrumental in human evolution and survival, enhancing the ability to locate food, resources, and water; identify danger; facilitate visual access; foster mobility; and more. Natural colors, such as earth tones, are thus often used to good effect by designers. | 環境色彩、自然色彩、大地色彩 | V |
| | 2. Water | Water features constantly elicit especially high levels of liking or preference (Ulrich, 1993). The effective use of water as a design feature is complex, and often contingent on such considerations as perceptions of quality, quantity, movement, clarity, and other characteristics. | 喜好、偏好、水的質量、數量、清晰度、引發感官知覺 | V |
| | 3. Air | People prefer natural ventilation over processed and stagnant air. Important conditions include quality, movement, flow, stimulation of other senses such as feel and smell, and visual appeal despite the seeming invisibility of the atmosphere. | 自然通風、品質、動態、流動、刺激感官 | V |
| | 4. Sunlight | Daylight is consistently identified as an important and preferred feature by most people in the built environment. | 陽光、舒適感 | V |
| | 5. Plants | Plants are fundamental to human existence as sources of food, fiber, fodder, and other aspects of sustenance and security. The mere insertion of plants into the built environment can enhance comfort, satisfaction, well-being, and performance. | 植物、生存基礎，提高舒適、滿意、幸福 | V |

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|---|---|---|---|---|
| 1. 環境特徵 (Environmental features) | 6. Animals | Animals are similarly basic to human existence as sources of food, resources, protection, companionship, and occasionally as precipitators of fear and danger. The presence of animal forms, nonetheless, often provokes satisfaction, pleasure, stimulation, and emotional interest. | 動物，食物、資源、保護、陪伴、動物形滿足、愉悅、情感興趣 | V |
| | 7. Natural materials | People generally prefer natural over artificial materials, even when the artificial forms are close or seeming exact copies of natural products. | 天然材料、<br>非人造材料 | V |
| | 8. Views and vistas | People express a strong and consistent preference for exterior views, especially when the vistas contain natural features and vegetation. These views are often most satisfying when the scale is compatible with human experience. | 偏好遠景、自然特徵與植被、尺寸適當、滿意 | V |
| | 9. Facade greening | Buildings with vegetative façades, such as ivy walls or green roofs, often provoke interest and satisfaction. | 植物立面建物，例如常春藤、興趣、滿意度 | V |
| | 10. Geology and landscape | The compatible connection of buildings to prominent geological features is often an effective design strategy. | 建築物與景觀環境 | V |
| | 11. Habitats and ecosystems | Buildings and landscapes that possess a close and compatible relationship to local habitats and ecosystems also tend to be highly effective and preferred. Important ecosystems in this regard are often wetlands, forests, grasslands, and watersheds. | 建物和景觀與當地生態棲地 | V |
| | 12. Fire | Fire in the built environment, while a complicated and difficult design challenge, is often a preferred feature, generally associated with the benefits of heating and cooking. The manipulated experience of fire within building interiors has long been celebrated as a sign of comfort and civilization, providing pleasing qualities of color, warmth, and movement. | 取暖 烹飪 建築內部的火溫暖、文明、屬人造物設計元素 | X |

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| 2. 自然形狀與樣式 (Natural shapes and forms) | 1. Botanical motifs | The shapes, forms, and patterns of plants and other vegetative matter are a frequent and often important design element of the built environment (Hersey, 1999). These representations often mimic or simulate plant forms such as foliage, ferns, cones, shrubs, and bushes, both literally and metaphorically. | 呈現或模擬「自然樣貌」之設計，讓整體環境增添植物形狀、樣式之自然體驗 | V |
| | 2. Tree and columnar supports | The appearance or simulation of tree-like shapes, especially columnar supports, is a common and often coveted design feature in the built environment. | 呈現或模擬「自然樣貌」之設計，讓整體環境增添樹形，如圓柱狀支撐的外觀設計 | V |
| | 3. Animal (mainly vertebrate) motifs | The simulation of animal life is widespread in building interiors and facades, although to a less extent than with plants. The appearance of animal parts is often encountered, such as claws, heads, rather than entire creatures. | 概念中指出在建築物的內部和外牆模擬動物型式 | X |
| | 4. Shells and spirals | Simulations and depictions of invertebrate creatures are widespread design features in the built environment. The shapes and forms of bees (and their hives), flies, butterflies, moths, and other insects, as well as spiders (and their webs) and other invertebrates, are also common. Some building designs mimic invertebrate processes. | 雖有提及建成環境，但偏重於建築物、設施物設計特徵：模仿無脊椎動物，如蜜蜂、蝶、昆蟲等的樣貌 | X |
| | 5. Egg, oval, and tubular forms | Egglike and tubular forms are also design elements in some building interiors, facades, and exterior landscapes such as gardens and fountains. These shapes often occur literally and metaphorically, both important expressions of ornament and sometimes for structural purposes. | 概念中指出運用蛋形或管狀設計於建築物的內部、外牆、外部景觀，例如：花園、水池等 | V |
| | 6.Arches, vaults, domes | Arches, vaults, and domes in the built environment resemble or copy forms found in nature, including beehives, nest-like structures, shell forms, and cliffs. These forms can be used for both decorative and sometimes for structural purposes. | 拱門、拱頂和圓頂設計、複製自然形式、裝飾等，偏重建築物或設施物設計 | X |

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| 2. 自然形狀與樣式 (Natural shapes and forms) | 7. Shapes resisting straight lines and right angles | Natural shapes and forms are often sinuous, flowing, and adaptive in responding to forces and pressures found in nature. Natural features are thus rarely revealed as straight lines and right angles characteristic of human engineering and manufactured products and structures. | 呈現自然的形狀與形式，如：蜿蜒、曲折的線條 | V |
| | 8. Simulation of natural features | This attribute reaffirms the tendency to simulate rather than replicate actual natural forms in the built environment. Ornamentation and decoration especially employ imagined forms only vaguely reminiscent of those found in the natural world. The designs are successful when they possess a logic that intimates functional features occurring in nature, which suggest structural integrity and adaptive advantage in response to environmental pressures rather than mere superficial decoration. | 偏重裝飾品、圖騰、形狀，讓人聯想到模擬自然設計 | X |
| | 9. Biomorphy | Some interesting architectural forms bear very little resemblance to life forms encountered in nature, yet are viewed as organic, for example, Sydney Opera House and the “pregnant whale” in Yale University. | 偏重建築物形式、生物形態、生物形態建築、屬建築物或設施物設計 | X |
| | 10. Geomorphology | Some building designs mimic or metaphorically embrace landscape and geology in relative proximity to the structure. This relationship to the ground can lend the appearance of solidity to the built environment, making structures appear integral rather than separate from their geological context. | 雖提到建築物設計，但提及建物設計與近景之融合度 | V |
| | 11. Biomimicry | Some successful designs borrow from adaptations functionally found in nature, particularly among other species, for example, include the structural strength and bioclimatic properties of shells, crystals, webs, mounds, and hives, effectively incorporated into the built environment. | 概念上廣泛指出藉由功能上的仿生性設計，能融入整體景觀環境中之氛圍 | V |

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| 3. 自然形式與過程 (Nature pattern and process) | 1. **Sensory variability** | Human fitness and survival have always required coping with a highly sensuous and variable natural environment, particularly responding to light, sound, touch, smell, and other sensory environmental conditions. Human satisfaction and well-being continue to be reliant on perceiving and responding to sensory variability, especially when this occurs in structured and organized ways within the built environment. | 運用感官知覺感受環境刺激如光線、聲音、自然元素等<br>**評論：感官知覺體驗環境，調至「環境與人互動關係」** | V |
| | 2. **Information richness** | The cognitive richness of the natural world reflects it is likely being the most intellectually challenging environment people will ever encounter even in our modern information age, which stimulates curiosity, imagination, exploration, discovery, and problem-solving. | 環境提供訊息，引發好奇心、想像力、探索等<br>**評論：環境資訊產生的心理反應，調至「環境與人互動關係」** | V |
| | 3. Age, change, and the patina of time | A fundamental feature of the natural world is aging through time, particularly organic forms, which evoke familiarity and satisfaction. A patina of time is characteristic of natural materials, even inorganic ones, and is one reason, as noted above, that artificial products rarely evoke sustained positive response even when they exact copies. | 自然界的特質會隨著時間的流逝、衰老等變化，而人造物是無法引發正向反應 | V |
| | 4.Growth and efflorescence | Growth and development are specific expressions of aging that when found in the built environment typically provoke pleasure and satisfaction. | 自然生長、發育、衰老等現象、引發愉悅和滿足 | V |
| | 5. Central focal point | The navigability of natural landscapes is often enhanced by the presence of a centrally perceived focal point. This point transforms what otherwise is a chaotic setting into an organized one that facilitates passage and way-findings. | 集中、焦點、環境有組織性 | V |
| | 6. Patterned wholes | People respond positively to natural and built environments when variability has been united by integrated and patterned wholes, and foster understanding and feelings mastery and control. | 環境有規律具整體統一性，促進理解與掌握度 | V |

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| 3. 自然形式與過程 (Nature pattern and process) | 7. Bounded spaces | Humans have a strong proclivity for bounded spaces. This territorial tendency, over evolutionary time, likely fostered resource exploitation and security, which enhance the clear recognition and consistent boundaries and place demarcations. | 環境中具清楚界線、邊界，提升辨識度 | V |
| | 8. Transitional spaces | Transitional spaces within and between built and natural environments often foster comfort by providing access from one area to another. | 過渡區、緩衝區、空間轉換 | V |
| | 9. **Linked series and chains** | Clear physical and temporal movement in both natural and built environments is often facilitated by linked spaces, especially when occurring in connected chains. These relational spaces covey meaning and organization, as well as sometimes a sense of mystery that both stimulates and entices. | 連結、意義、組織、神秘<br>**評論：空間整體產生的心理反應，調至「環境與人互動關係」** | V |
| | 10. Integration of parts to wholes | People prefer in natural and built environments the feeling that discrete parts comprise an overall whole, particularly when the whole is an emergent property consisting of more than the sum of the individual parts. | 自然和建築組成一個整體的感覺 | V |
| | 11. Complementary contrasts | Meaning and intelligibility, as well as interest and stimulation, in natural and constructed settings, often reveal the blending of contrasting features in a complementary fashion. | 對比性（明亮、高低、開放與封閉） | V |
| | 12. Dynamic balance and tension | The dynamic balancing of different and sometimes contrasting forms often fosters a sense of strength and durability in both natural and built environments. | 強調無機體與有機體的動態平衡協調 | V |
| | 13. Fractals | These structures frequently include repeated but varying basic designs, such as ornamentation in parallel or closely linked rows that differ slightly from one another. | 序變化、重複性 | V |
| | 14. Hierarchically organized ratios and scales | Successful natural and built forms often occur in hierarchically connected ways, sometimes arithmetically or geometrically related. | 自然形式和構造形式通常以層次連接、幾何、比例 | V |

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| 4. 光與空間 (Light and space) | 1. Natural light | This attribute includes the effects of daylighting as previously described, as well as the inclusion of the full-color spectrum of natural light. | 日光照明的效果及是否有自然光的色譜，與Sunlight概念相似，故合併該念至同一個問項內 | X |
| | 2. Filtered and diffused light | Filtered or diffused sunlight can also stimulate observation and feelings of connection by providing a variable and mediated connection between spaces, particularly inside and outside areas. | 建成空間內外光線的連結、重視「光」在建築物體的串聯效果的設計，而非單指戶外綠地環境 | X |
| | 3. **Light and shadow** | The complementary contrast of light and dark spaces can produce significant satisfaction in both buildings and landscapes. | 涉及景觀環境之光影對比性設計<br>**評論：空間配置中光影的對比性，調至「自然形式與過程」** | V |
| | 4. Reflected light | Lighting designs are frequently enhanced by light reflecting off surfaces such as light-colored walls, ceilings, and reflective bodies like water. Functional benefits include mitigation of glare, enhanced penetration of light into interior spaces, and spying resources at a distance. | 建成空間內的環境是否有自然、反射光、減輕眩光、增強光線在內部空間中的穿透感 | X |
| | 5. Light pools | People are often drawn into and through interior spaces by the presence of pools of connected light. Light pools can assist movement and way-finding by providing lighted patches across shadowed or obscured areas such as a forest or darkened halls and passageways. | 光線吸引室內空間 | X |
| | 6. Warm light | The perception of warmly lit areas, often islands of modulated sunlight surrounded by darker spaces, can enhance the feeling of a nested, secure, and inviting interior. | 室內空間的光、安全感 | X |
| | 7. Light as shape and form | The manipulation of natural light can create stimulating, dynamic, and sculptural forms. | 屬於人為設計操作自然光的手法 | X |

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| 4. 光與空間 (Light and space) | 8. **Spaciousness** | People prefer feelings of openness in natural and built environments, especially when it occurs in complementary relation to sheltered protected refuges at the surrounding edges. Effective designs often include spacious settings in close alliance with smaller spaces. | 開闊、寬敞空間包含小空間<br>**評論：空間配置，調至「自然形式與過程」** | V |
| | 9. **Spatial variability** | Spatial variability fosters emotional and intellectual stimulation. Spatial diversity is often most effective when in complementary relation to organized and united spaces. | 空間變異性、刺激情感或智力<br>**評論：空間與人產生的心理反應，調至「環境與人互動關係」** | V |
| | 10. Space as shape and form | Space can be creatively manipulated to convey shapes and forms. This effect can add beauty to the built environment. | 重視人為操縱空間設計的手法，創造美感、引法好奇心、探索與發現 | X |
| | 11. Spatial harmony | The manipulation of space in the built environment tends to be most effective when it blends light, mass, and scale within a bounded context. | 在有限空間內，進行人造光、質量與尺寸的設計操作 | X |
| | 12. Inside-outside spaces | Appealing interior spaces in the built environment often appear connected to the outside environment. These areas also mark the transition of nature with culture. Important design forms in the built environment that evoke this quality include colonnades, porches, foyers, atriums, and interior gardens. | 將自然元素帶入建成空間設計 | X |

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| 5. 地方感與連結的關係 (Place-based relationship) | 1. Geographic connection to place | Secure feelings of connection to the geography of an area often foster feelings of familiarity and predictability. This can be achieved by emphasizing prominent geological features associated with the siting, orientation, and views of buildings and landscapes. | 地理環境與建築物的連結 | V |
| | 2. Historic connection to place | Meaningful relation to place often marks the passage of time, which fosters a sense of participation and awareness of an area's culture and collective memory. Buildings and landscapes that elicit this continuity with the past encourage the belief that the present and future are meaningfully linked to the history of a place. | 與地方意義的聯結、區域文化、集體記憶的 | V |
| | 3. Ecological connection to place | Places are sustained by an affirmative connection to ecology, particularly prominent ecosystems such as watersheds and dominant biogeographical features (e.g., mountains, deserts, estuaries, rivers, and oceans). | 與生態系統的聯繫、地理特徵，例如，山脈，沙漠，河口，河流和海洋 | V |
| | 4. Cultural connection to place | Cultural connection to place integrates the history, geography, and ecology of an area, becoming an integral component of individual and collective identity. | 與地方的文化連結、融合地區的歷史、地理和生態、認同感 | V |
| | 5. Indigenous materials | A positive relation to place is generally enhanced by the utilization of local and indigenous materials. Native resources provide a vivid and resonant reminder of local culture and environment, as well as less energy for manufacture and transport. | 使用當地素材 | V |
| | 6. Landscape orientation | Buildings and landscapes that compatibly connect to the local environment contribute to a sense of place. The orientation to landscape frequently evokes a sense of being a part of and embedded within local settings, rather than being separated from them. | 與當地環境相容性、營造一種地方感 | V |

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| 5. 地方感與連結的關係 (Place-based relationship) | 7. Landscape features that define building form | Landscape features can embellish and distinguish building forms, particularly prominent geological features, natural objects, and water. The built environment can integrate with rather than be isolated from its biophysical context. | 建成環境特徵與其生物物理環境融合而不孤立 | V |
| | 8. **Landscape ecology** | Effective place-based designs reinforce landscape ecology over the long term, which considers landscape structure, pattern, and processes such as ecological connectivity, biological corridors, resource flows, biodiversity, optimal scale and size, ecological boundaries, and other parameters of functioning natural systems. | 生態系統設計手法，考慮景觀結構、格局、過程<br>**評論：生態空設計，調至「自然形式與過程」** | V |
| | 9. Integration of culture and ecology | The fusion of culture with ecology fosters long-term sustainability. The result marks the point where nature and humanity are positively transformed and mutually enriched by their association. | 文化與生態融合、可持續發展 | V |
| | 10. Spirit of place | The spirit of a place signifies a level of commitment and meaning that people extend to both natural and built environments when they become cherished components of individual and collective identity, more than simply inanimate matter. The spirit of place metaphorically signifies the built environment having become life-like and serving as the motivational basis for long-term stewardship and responsibility. | 地方精神象徵、延伸到無生命的事物與自然和建成環境 | V |
| | 11. Avoiding placelessness | "Placelessness" is the antithesis of place-based design and to be avoided whenever possible. This corrosive separation of the built environment from its bio-cultural context has resulted in the decline of human-nature relationships and environmental sustainability. | 脫離地方文化或生態的聯繫的設計 | V |

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| 6. 環境與人互動關係 (Evolved Human-Nature interaction) | 1. Prospect and refuge | Refuge reflects a structure or natural environment's ability to provide a secure and protected setting. Prospect, on the other hand, emphasizes discerning distant objects, habitats, and horizons, evolutionarily instrumental in locating resources, facilitating movement, and identifying sources of danger. | 避難、建築物或自然環境提供安全、保護、前景、辨別遠處的物體、識別危險 | V |
| | 2. **Order and complexity** | Order is achieved in the built or natural environment by imposing structure and organization. Extreme order often results in repetition, monotony, and boredom. Complexity reflects the occurrence of detail and variability. Excessive complexity makes it difficult to assimilate detail and sometimes leading to a sense of chaos. | 秩序、重複、單調、複雜性、細節、可變性、混亂<br>**評論：空間設計手法，調至「自然形式與過程」** | V |
| | 3. Curiosity and enticement | Curiosity reflects the human need for exploration, discovery, mystery, and creativity, all instrumental in problem-solving (Kaplan, Kaplan, & Ryan, 1998). Enticement fosters curiosity. | 好奇心、探索、發現、神秘、創造力 | V |
| | 4. **Change and metamorphosis** | Change is a constant in both natural and human systems, reflected in the processes of growth, maturation, and metamorphosis (Bloomer, 2000). | 變化、生長<br>**評論：環境變化，調至「自然形式與過程」** | V |
| | 5.Security and protection | A fundamental objective of the built environment is ensuring protection from threatening forces in nature. Security in the built environment must not excessively insulate or isolate people from the natural world. | 保護、安全 | V |
| | 6. Mastery and control | Buildings and constructed landscapes reflect the human desire for mastery and control over nature. When accomplished with moderation and respect, mastering nature facilitates the satisfactory expression of human ingenuity and cleverness that fosters self-confidence and self-esteem. | 建築物和人工景觀反映人類對自然的掌握和控制 | V |

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| 6. 環境與人互動關係 (Evolved Human-Nature interaction) | 7. Affection and attachment | Affection for the natural world has been a critical component in engendering the human capacities for bonding and attachment, important in a largely social creature. Buildings and landscapes elicit strong emotional affinities for nature are typically recipients of lasting loyalty and commitment. | 對自然偏好、依戀、強烈情感親合度 | V |
| | 8. Attraction and beauty | The aesthetic attraction to nature is one of the strongest inclinations of the human species. This biologically encoded tendency has been instrumental in fostering the capacities for curiosity, imagination, creativity, exploration, and problem-solving. | 審美、吸引力、培養好奇心、想像力、創造力、解決問題能力 | V |
| | 9. Exploration and discovery | Nature is the most information-rich and intellectually stimulating environment that people ever encounter. | 大自然很多訊息、探索、發現 | V |
| | 10. Information and cognition | Intellectual satisfaction and cognitive prowess can be fostered through designs that emphasize the complexity of natural shapes and forms. It can be achieved through the direct and indirect experience of nature, as well as by the creative use of ornamentation in the built environment that fosters critical thinking and problem-solving. | 通過自然形狀和形式的複雜性的設計，培養智力上的滿足感、認知能力滿足感、解決問題能力 | V |
| | 11. Fear and awe | Fear of nature can also be a motivational basis for designing peril and adventure into the built environment. Feelings of awe for the natural world can further combine reverence with fear. | 恐懼、厭惡、敬畏 | V |
| | 12. Reverence and spirituality | Some of the cherished building designs affirm the human need for establishing meaningful relation to creation. These designs provoke feelings of transcendence and enduring connection that defy the aloneness of a single person isolated in space and time. Structures that achieve this reverential feeling of connection are also typically sustained generation after generation. | 建成環境或建物空間具有意義的關係，產生超越體驗 | V |

備註:以粗體字表示調整部

表2 戶外建成環境之感知親生命性設計問項

| | 環境特徵及元素<br>Environmental attributes and elements | | 問項<br>Items |
|---|---|---|---|
| 1. 環境特徵 (Environmental features) | 1. 顏色<br>Color | 1. | 這裡的環境有自然色彩(例如:大地色)。<br>There are natural colors here (e.g., earth tones). |
| | 2. 水<br>Water | 2. | 這裡的環境有水。<br>There is water here. |
| | 3. 空氣<br>Air | 3. | 這裡的環境空氣很好。<br>The ambient air is good here. |
| | 4. 陽光<br>Sunlight | 4. | 這裡的環境有陽光、自然光線。<br>There are sunlight and natural light here. |
| | 5. 植物<br>Plants | 5. | 這裡的環境有植栽。<br>There are plants here. |
| | 6. 動物<br>Animals | 6. | 這裡的環境有動物。<br>There are animals here. |
| | 7. 自然材質<br>Natural materials | 7. | 這裡的環境有自然材質的設施(例如:木平台)。<br>There are facilities made of natural materials here (e.g., wooden platforms). |
| | 8. 景觀<br>Views and vistas | 8. | 這裡的環境可以看到自然景觀。<br>There is a natural landscape here. |
| | 9. 立面綠化<br>Facade greening | 9. | 這裡的環境有立面綠化(例如:爬藤植物、植生牆)<br>There is facade greenery here (e.g., vine, vertical garden). |
| | 10. 地理與景觀<br>Geology and landscape | 10. | 這裡的建築物與該環境的地景特徵相融合。<br>The buildings here blend with the landscape features. |
| | 11. 棲息地與生態系<br>Habitats and ecosystems | 11. | 這裡的建築物、景觀融合於生態棲地環境中。<br>The buildings and landscapes here are integrated into the ecological habitat. |
| 2. 自然形狀與樣式 (Natural shapes and forms) | 1. 植物圖案<br>Botanical motifs. | 12. | 這裡的環境有模擬植物型態的設計(例如:葉形)。<br>There are designs that simulate the shape of plants here (e.g., leaves shape). |
| | 2. 樹狀和柱狀<br>Tree and columnar supports | 13. | 這裡的環境有模擬樹型外觀的設計(例如:樹幹)。<br>There are designs that simulate the appearance of trees here (e.g., trunks). |
| | 3. 卵，卵形和管狀<br>Egg, oval, and tubular forms | 14. | 這裡的環境有卵形、橢圓形或管狀設計(例如:花園、噴泉)。<br>There are oval, elliptical, or tubular designs here (e.g., gardens, fountains). |
| | 4. 非直線或直角形狀<br>Shapes resisting straight lines and right angles | 15. | 這裡的環境有自然線條(例如:蜿蜒步道)。<br>There are natural lines here (e.g., trails). |
| | 5. 地貌學<br>Geomorphology | 16. | 這裡的建築物設計與近景融合構成完整地景(例如:社科院、博雅館)。<br>The buildings are integrated into the foreground and form a complete landscape (e.g., College of Social Sciences Building, Liberal Education Classroom Building). |
| | 6. 仿生學<br>Biomimicry | 17. | 這裡的環境有模仿生物型態的設計。<br>There are designs that imitate the biological forms in general here. |

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| | 環境特徵及元素<br>Environmental attributes and elements | 問項<br>Items |
|---|---|---|
| 3. 自然形式&過程 (Nature pattern and process) | 1. 年齡、變化和時間<br>Age, change, and the patina of time | 18. 這裡的環境能看到時間的推移或老化的現象(例如:樹葉凋落、風化作用)。<br>There is a passage of time or phenomenon of aging here (e.g., leaf falling, weathering). |
| | 2. 生長和風化<br>Growth and efflorescence | 19. 這裡的環境能看到自然的生長現象。<br>There is natural growth here. |
| | 3. 變化和變態<br>Change and metamorphosis | 20. 這裡的環境具有生長、成熟、季節等變化的過程。<br>There is a process of growth, maturity, and seasonal change here. |
| | 4. 動態平衡和張力<br>Dynamic balance and tension | 21. 這裡的人造結構物、設施(無機體)與自然(有機體)搭配得宜。<br>The man-made structures and facilities (inorganic) and nature (organic) match well here. |
| | 5. 景觀生態<br>Landscape ecology | 22. 這裡的環境採用生態廊道、生物多樣性等生態設計手法(例如:水圳)。<br>There are ecological design methods, such as corridors and biodiversity here (e.g., water pathway). |
| | 6. 焦點<br>Central focal point | 23. 這裡的建成環境有視覺焦點。<br>There is a central focal point here. |
| | 7. 整體<br>Patterned wholes | 24. 這裡的環境有整體性。<br>There are patterned wholes here. |
| | 8. 互補對比<br>Complementary contrasts | 25. 這裡的環境有對比性(例如:明/暗、高/低、開放/封閉)。<br>There are complementary contrasts (e.g., light/dark, high/low, open/closed) here. |
| | 9. 光和影<br>Light and shadow | 26. 這裡的環境有光影的對比性，並可分辨距離感。<br>There are contrasts of light and shadow and a sense of distance can be distinguished here. |
| | 10. 碎形<br>Fractals | 27. 這裡的環境有重複性及變化性(例如:花圃、植栽種類)。<br>There are repetitive and variable structure designs here (e.g., flower beds, planting types). |
| | 11. 分層組織的比率和比例<br>Hierarchically organized ratios and scales | 28. 這裡的環境有層次感及適當的尺寸比例。<br>There are senses of hierarchy and appropriate size ratios here. |
| | 12. 秩序和復雜性<br>Order and complexity | 29. 這裡的環境組成有秩序且多樣性。<br>The composition is orderly and diverse here. |
| | 13. 明顯界線空間<br>Bounded spaces | 30. 這裡的環境與外在環境有明顯的區隔界線。<br>The environment is separated from the external environment here. |
| | 14. 過渡空間<br>Transitional spaces | 31. 這裡的環境有自然元素作為空間轉換。<br>There are natural elements as a spatial transformation here. |
| | 15. 整合一體<br>Integration of parts to wholes | 32. 這裡的環境元素各自能組成一個空間。<br>The elements can integrate parts of a whole. |
| | 16. 開闊空間<br>Spaciousness | 33. 這裡的環境具有開闊性及視覺延伸感。<br>There are senses of openness and visual extension here. |

| 環境特徵及元素<br>Environmental attributes and elements | | 問項<br>Items |
|---|---|---|
| 4. 地方感與連結的關係 (Place-based relationship) | 1. 與地理性連結<br>Geographic connection to place | 34. 這裡的環境能讓我與該地景的地理上的連結並產生熟悉感(例如：芳蘭山、蟾蜍山、101 大樓)。<br>The environment allows me to connect with the geographical location and generate a sense of familiarity (e.g., Fanglan Mountain, Toad Mountain, 101Building). |
| | 2. 與歷史性連結<br>Historic connection to place | 35. 這裡的環境能讓我與該地景產生歷史上的連結感並引發親生經歷的感覺(例如：十三溝面磚、傅鍾、鐘聲)。<br>The environment allows me to have a sense of historical connection with the landscape and triggers the feeling of personal experience (e.g., 13-Groove Tile, NTU Bell Fu, bells sound). |
| | 3. 與生態性連結<br>Ecological connection to place | 36. 這裡的環境能讓我與整體景觀及生態系統產生連結感(例如:醉月湖、生態池)。<br>The environment allows me to have a sense of connection with the overall landscape and ecosystem (e.g., Drunken Moon Lake, ecological pond). |
| | 4. 與文化性連結<br>Cultural connection to place | 37. 這裡的環境能讓我與該地景的文化產生認同感(例如：杜鵑花節)。<br>The environment allows me to identify with the culture (e.g., Rhododendron Festival). |
| | 5. 當地材料<br>Indigenous materials | 38. 這裡的環境運用當地素材，增加與該地的連結感。<br>There are indigenous materials to increase a sense of connection with the place. |
| | 6. 景觀定向<br>Landscape orientation. | 39. 這裡的環境與建築物相融合營造當地地方感(例如:椰林大道)。<br>The environment and buildings are integrated to create a sense of place (e.g., Palm Avenue). |
| | 7. 景觀特徵與建築形式<br>Landscape features that define building form | 40. 這裡的環境融入外圍都市景觀。<br>The environment blends into the surrounding urban landscape. |
| | 8. 文化與生態融合<br>Integration of culture and ecology | 41. 這裡的人文與自然環境具有永續發展的特質。<br>The culture and ecology integrate a sense of sustainable development. |
| | 9. 地方精神<br>Spirit of place | 42. 這裡的環境具有地方象徵意義。<br>There is a spirit of place. |
| | 10. 避免無地方意義<br>Avoiding placelessness | 43. 這裡的環境不具地方意義，無法產生連結感。<br>There are placelessness and no sense of connection. |
| 5.環境與人互動關係 (Evolved Human-nature interaction) | 1. 前景與避難<br>Prospect and refuge. | 44. 這裡的環境能眺望遠處、同時具有遮蔽及保護的功能。<br>There is a sense of prospect and refuge here. |
| | 2. 掌握與控制<br>Mastery and control | 45. 這裡的環境有種人類主宰空間的感覺。<br>There is a sense of human domination and control here. |
| | 3. 感官豐富<br>Sensory variability | 46. 這裡的環境有自然元素能豐富感官體驗。<br>Natural elements are enriching the sensory experience here. |
| | 4. 訊息豐富<br>Information richness | 47. 這裡的環境元素豐富能刺激想像力。<br>The richness in environmental information stimulates imagination here. |
| | 5. 探索與發現<br>Exploration and discovery | 48. 這裡的環境值得我去探索與發現。<br>The environment is worth exploring and discovering here. |

| | 環境特徵及元素<br>Environmental attributes and elements | 問項<br>Items |
|---|---|---|
| 5.環境與人互動關係 (Evolved Human-nature interaction) | 6. 好奇與誘惑<br>Curiosity and enticement | 49. 這裡的環境能引發好奇心。<br>The environment arouses curiosity here. |
| | 7. 安全與保護<br>Security and protection | 50. 這裡的環境能提供安全感。<br>The environment provides a sense of security here. |
| | 8. 訊息與認知<br>Information and cognition | 51. 這裡的環境能引發滿意感。<br>The environment triggers a sense of satisfaction here. |
| | | 52. 這裡的環境能引發思考及解決問題能力。<br>The environment triggers thinking and problem-solving skills here. |
| | 9. 連結<br>Linked series and chains | 53. 這裡的自然與建成環境能引發神秘感。<br>The natural and built environments trigger a sense of mystery here. |
| | 10.吸引力與美麗<br>Attraction and beauty | 54. 這裡的環境很美麗，能吸引我。<br>The environment is beautiful and attractive to me. |
| | 11.情感與依附<br>Affection and attachment | 55. 這裡的環境能引發對自然強烈的依附感。<br>The environment triggers a strong sense of attachment to nature. |
| | 12.空間變化<br>Spatial variability | 56. 這裡的環境具有變化性，能引發情緒。<br>The environment is a spatial variable and triggers emotions. |
| | 13.恐懼與敬畏<br>Fear and awe | 57. *這裡的環境讓我感到害怕。<br>The environment makes me feel afraid. |
| | | 58. 這裡的環境讓我感到敬畏。<br>The environment makes me feel awe. |
| | 14.敬畏與靈性<br>Reverence and spirituality | 59. 這裡的環境讓我沉浸於空間與時間中。<br>The environment makes me feel immersed in space and time. |

備註：1. 若問項後方有舉例者，可依照實驗環境特色做調整。2. *反向題。

## 三、未來應用與發展

儘管眾多環境心理學相關的研究及環境設計，都試圖將「自然」融入日常生活環境，然而如何在都市或綠色環境中感受到親生命性設計，拉近人與自然互動的關係，進而增進生心理健康，是一項能重要的環境、心理與健康之課題。親生命及親生命性的概念涉及到不少環境心理學理論與景觀設計的手法，本研究採用親生命性設計的概念作為發展測量項目的依據，從各項環境屬性、特徵與元素的定義，萃取關鍵詞並重新整合環境屬性、特徵與元素，最後擬定出59項評估戶外建成環境親生命性設計的問項架構，包含五項環境屬性，分別為(1)環境特徵(Environmental features)、(2)自然形狀與樣式(Natural shapes and forms)、(3)自然形式與變化過程(Natural patterns and processes)、 (4)地方感與連結的關係(Place-based relationship)與 (5)環境與人互動關係(Evolved human-nature relationship)；前三項屬性為評估環境中的特徵與元素，後兩項屬性評估人與環境互動時產生之心理反應。就未來應用方面，建議採用李克特五點尺度讓受測者回答「感知親生命性設計」問項，並進一步以統計檢定驗證項目之信效度，作為以親生命性觀點（西方環境觀）輔助了解都市環境中感知環境屬性、特徵、元素與人互動之科學化評估量表。

傳統環境氣(TEQ)是從東方環境觀點探討環境組成與佈局的特徵，涵蓋人與環境互動間產生平衡、和諧與整體感；親生命性設計重視細微的環境特徵與元素及其產生人與環境互動之心理反應。不論從東方或西方的環境觀點切入，皆重視人們感知環境特徵或元素後，增進生心理效益與健康。因此就環境屬性、特徵與元素而言，可分別從東方環境觀與西方環境觀論述其差異性外，也可進一步針對兩種量表間的差異做統計檢定。另一方面，未來相關研究（如圖2）除了不斷檢視感知親生命性設計量表的穩定性，另可配合張俊彥(2019)開發之健康雲端系統(HealthCloud)，大數據收集與調查人們感知當下環境之親生命性設計屬性、特徵與元素與心跳值後，建立「感知親生命性生心理反應資料庫」，更能深入探討環境類型或族群對所重視的環境特徵與元素，作為未來景觀規劃與設計健康環境時之科學化數據參考。

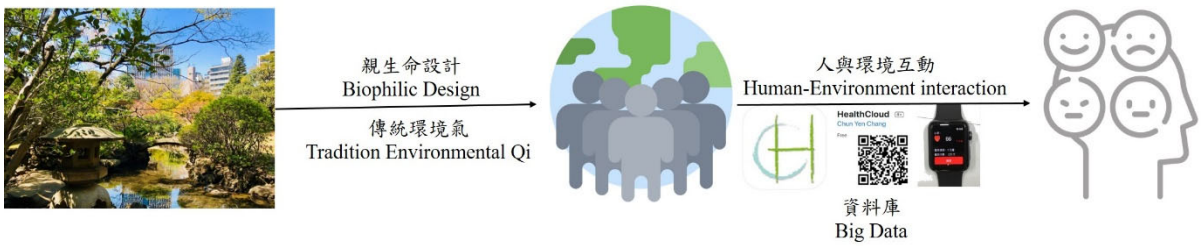


圖2　親生命性設計與傳統環境氣對人與環境效益之概念圖

## 肆、結論

都市環境人口快速地增長，都市綠地又逐漸被鋼筋混凝土與大量建築物體所包圍，無形中減少人們與自然環境或元素等接觸經驗。都市環境中的各種自然或綠色設計，如街道上的行道樹、雨水花園、綠屋頂、草地或都市農園等眾多環境設計手法，可改善人們的生心理健康與福祉。另一方面，近年來聯合國提出全球永續發展目標(SDGs)，如目標3：良好的健康與福祉(Good health and well-being)、目標11：可持續的都市和社區(Sustainable cities and communities)以及目標15：陸上生活(Life on Land)等，與永續設計目標與設計手法有關，皆能隱約連結到親生命與親生命性設計之議題與概念。本研究著重在建構親生命性設計之概念轉化成可被測量之項目，進而探討人們感知到環境中重要之親生命性屬性、特徵或元素及其與環境互動之體驗。未來建構之量表對學術研究人員、實務環境相關規劃與設計師及政府單位等，亦是值得注意與採取改善都市環境的行動與解決環境相關議題之方案。

就學術研究貢獻方面，本篇研究目的在建構感知親生命性設計項目，真實測量人類所感受到且會影響環境體驗及其引發生心理效益的環境特徵與元素，藉此能進一步連結過去東西方環境觀點之相關理論。舉例來說，藉由東方傳統環境氣的概念，了解人們重視景觀空間結構的尺寸大小、景觀整體性、自然曲線、自然元素、植栽配置、光線、視野、微氣候因子等環境特徵及其產生之生心理健康感受（例如：思緒沉澱、不緊張、自在、身體舒暢、有活力、放鬆等等），同時思考環境心理學相關理論與篩選親生命性設計的特徵與元素，作為詮釋親生命性設計內涵的項目。除此之外，親生命性設計項目分為五種環境屬性，前三項屬性分別為環境特徵、自然形狀與樣式及自然形式與過程，除了提及環境特徵與屬性外，也涉及到眾多景觀空間的設計手法，例如重視元素間的互補、對比性、比例、複雜度、開闊性等，能做為未來實驗設計上挑選景觀刺激物的操作原則；後兩項屬性是地方感與連結的關係及環境與人互動關係，除了能討論地方連結的相關理論與實證研究外，更能夠將Kaplan(1987)及Kaplan與Kaplan(1989)提出的相關理論作討論。另一方面，未來研究不論在文獻回顧的論述上或者結果與討論部分，能運用親生命性設計的概念，深入淺出連結景觀偏好理論、眺曖理論、棲地理論、草原假說、恢復性環境特徵、景觀生態理論等，作為詮釋環境組成與佈局所涵蓋的特徵與元素，亦能提

升對建成環境與人們互動的價值。

本研究重點將親生命性設計之概念轉換成具體、可被測量之問項，藉此衡量人感知到環境之親生命性屬性、特徵、元素及其體驗。本研究從親生命性設計之概念意義切入主要涉及幾點因素：第一，因親生命性設計涉及許多抽象性概念，若採用質性訪談法，除了一般大眾可能無法清楚理解親生命性設計的定義及區分環境屬性與特徵外，相關環境與人互動之情感上用詞可能有限，會導致不容易精準地獲得親生命性設計之內涵；第二，從文獻回顧發現親生命性設計與環境心理學理論息息相關，鑒於相關理論與知識，有利於萃取各環境特徵與元素之用字與關鍵詞句，將概念具體化後並發展成感知親生命性設計項目。未來的研究便可推論與運用到建成環境類型、戶外綠地等，驗證該量表多樣化的信度與效度，且可深入比較感知東方環境觀（傳統環境氣）與西方環境觀（親生命性設計）之異同性，作為探討環境與人互動體驗對情緒、注意力恢復或偏好之影響。

就實務規劃與設計應用，除了可作為基地調查與設計發想之評估工具，結合健康雲端系統(HealthCloud)之大數據，可建立「具有親生命性設計意象之科學數據資料庫」，提供規劃設計師融入該項指標在環境設計的工作坊，建立業主、民眾、使用者與其他利益者對環境體驗與設計之有效溝通工具。另一方面，從一系列Kellert所撰寫的親生命性設計書籍中，列舉不少建成環境運用親生命性設計的環境特徵與元素，例如：落水山莊、京都平等院、雪梨歌劇院、新加坡濱海灣公園等，意喻這些實質環境場域佔有影響人與環境互動的健康效益。另從弗吉尼亞大學建築學院在網路平台建置目前具親生命性設計的合作城市(https://www.biophiliccities.org/partner-cities)，例如：新加坡、德奧斯丁、舊金山、聖路易斯、西班牙等等，可看出該網站探討如何落實綠色基礎設施、公園與綠地、垂直綠化等親生命性設計手法，強調串聯城市、花園、城市森林之綠色與生態網絡等，達到永續都市景觀設計作為具體拉近都市人們親近自然、獲得身心靈等健康效益的設計依據。

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